

Lesson 3-5 · Period 2

Multiplicity and Factored-Form Graphing

Klimsara-adapted · Student-centered · No Blooket

Do Now — Find the zeros from factored form

Algebra 2 · Lesson 3-5 · Period 2

DOK 1

5 min

bridge from Monday

Practice (3-5-savvas-q13) · Sketch from zeros Sketch the graph of the function by finding the zeros:

$$g(x) = (x + 3)(x - 1)(x - 4)$$

For each factor: what zero does it give? Mark each zero on your sketch.

Use the sentence frame: "The zeros are $x = \underline{\quad}$, $x = \underline{\quad}$, and $x = \underline{\quad}$ because each factor equals zero."

DOK 2

12 min

teacher models

Multiplicity Rules — keep on your page all period

1. If a factor appears **ODD** number of times → the graph **CROSSES** the x-axis at that zero.
2. If a factor appears **EVEN** number of times → the graph is **TANGENT** (touches and turns back) at that zero.

Example (Savvas Ex 2): $f(x) = (x - 2)^3(x + 1)^2$

- $x = 2$: factor appears 3 times — ODD → **crosses**
- $x = -1$: factor appears 2 times — EVEN → **tangent**

Explore — TPS (Think-Pair-Share) Algebra 2 · Lesson

3-5 · Period 2

DOK 2

33 min

student work

5 items in order. For each zero: write multiplicity + crosses/tangent before sketching. ~6–7 min each.

Try It 2a · Describe zeros $f(x) = x(x+4)(x-1)^4$ — describe graph behavior at each zero.

Try It 2b · Real zeros only $f(x) = (x^2 + 9)(x-1)^5(x+2)^2$ — which zeros are real? Crosses or tangent?

Practice #14 · Factor first $f(x) = x^3 + 6x^2 + 9x + 5$ — find

DOK 2

5 min

academic conversation + bridge prompt

Sentence frame · Pick one of your zeros “For $f(x) = \underline{\hspace{2cm}}$, the zero $x = \underline{\hspace{2cm}}$ has multiplicity $\underline{\hspace{2cm}}$, so the graph $\underline{\hspace{2cm}}$ the x-axis at that zero.”

Bridge to Wednesday (Period 3 Do Now) A cubic has zeros 2, $1 + \sqrt{3}$, and $1 - \sqrt{3}$. What is the nature of its coefficients — rational, irrational, or complex?

Exit Ticket — Summary Recap

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DOK 1-2

2 min

not graded

Complete on your exit ticket

1. Today I learned that _____.
2. The rule I want to remember is _____.
3. One thing I'm still unsure about is _____.